

Development Report

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Prepared by: Islem

Work Completed

1. Publeader — Validations workflow (AD1)

- Built the combined admin validation queue: drivers, companies, and partners share a single backend pipeline with per-kind detail, approve / reject / request-info actions, and outbound email notifications on every decision.
 - Each entity carries a per-kind review trail (last action only — reviewer, timestamp, rejection reason, last info request) so the queue UI can render history without a separate audit table.
 - Web validation queue page lists pending submissions and exposes the detail view with documents.
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2. Publeader — Finance & Invoicing (AD2)

- Built the invoice subsystem on top of the new admin finance area: CRUD with status machine (**brouillon** → **envoyée** → **payée**, derived **en retard** computed on read), atomic per-year invoice reference counter, per-line totals, configurable VAT.
 - Send-invoice route generates the PDF, emails the recipient via SMTP, and flips the invoice to **envoyée** in one transaction.
 - Driver commissions, internal expenses ledger (4 categories), and finance KPIs (MRR, collections, pending, commissions due, overdue) wired into a single admin finance page.
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3. Publeader — Stripe payments (AD3)

- Integrated Stripe Checkout (hosted) for invoice payment. Choices: hosted (PCI-light), email + advertiser web dashboard pay surface, refund/dispute reverts the invoice to **envoyée** with timestamp + reason flag.
- Persistent pay link strategy: each click on the email link mints a fresh Checkout Session via **/api/pay/[id]/redirect** so emails never expire even past Stripe's 24h Checkout TTL.
- Webhook handler with signature verification and idempotency (the Stripe event id is the primary key on a dedicated **stripe_events** collection; duplicate deliveries are ignored). Events handled: checkout completed, async payment succeeded, payment intent failed, charge refunded, dispute

created/closed (won/warning_closed clears the flag, lost is treated as a refund).

- Advertiser web endpoints for listing own invoices and creating Checkout Sessions on demand. Send-invoice email body now embeds a Pay button when Stripe is configured.

4. Publeader — Reports (AD4)

- Six report builders, all generated inline (sync) and uploaded to Cloudinary as `raw` resources, with persisted history per request:
 - **Bilan mensuel** (PDF) — revenue, commissions, expenses, net margin, active campaigns, top advertisers.
 - **Export comptable** (ZIP of three CSVs) — invoices, commissions, expenses, RFC-4180 quoted, UTF-8 BOM for Excel.
 - **Performance Leader Borne** (PDF) — per-terminal sprays, impressions, refills, partner revenue using the same rates as the partner statements.
 - **Activité chauffeurs** (PDF) — top performers by commissions over the period, capped at 100.
 - **Rapport annonceurs** (PDF) — campaigns, budget, impressions, fill rate (flocage drivers + borne impressions), invoiced + collected.
 - **Audit RGPD** (PDF) — PII inventory by collection with retention policy and live row counts. Sourced from a single retention-policy const so legal review has one document to maintain.
- Period picker per request (start/end). Admin web page replaces the old static mocks: builder card with 6 radio cards + date range, history grid with download/delete per entry.

5. Publeader — Dashboards & KPIs (AD5)

- Single admin dashboard endpoint runs ~9 parallel Mongo queries: finance KPIs (reusing AD2's service), counts (drivers, companies, partners, campaigns), borne fleet health (online/offline/maintenance from heartbeat freshness, monthly revenue, top terminals), city distribution, validation queue summary, recent campaigns, and month-over-month deltas on every headline metric.
- Revenue chart endpoint with hybrid data source: paid invoices for past days (attributed to flocage/borne via the linked campaign), accrued campaign budgets distributed linearly across active windows for today. Range selector supports 30 / 90 / 365 days.
- Advertiser dashboard endpoint: own impressions over the past 30 days with sparkline + delta vs prior 30, % toward `borne.targetImpressions` goal, active campaigns, billing summary (paid this month + open + overdue), recent campaign events as activity feed.
- Wired all three screens (`DashboardGlass`, `DashboardPro`, `EnterpriseDashboard`) to live data, replaced the hard-coded mock arrays and the synthetic stacked-area generator.

6. Publeader — Global search palette (AD6)

- Admin-only `/api/admin/search` with case-insensitive regex (escaped) over campaigns (brand/title/domain/city), drivers (name/phone/city), companies (name/contact/SIRET/VAT/city), and terminals (code/name/address/city). Per-type cap of 5, total 20 max.
 - 🏠K palette wired to live search with 250 ms debounce and an `AbortController` cancelling in-flight requests on every keystroke. Mock entities removed; `Résultats` group only renders for queries ≥ 2 characters.
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7. Publeader — User management & platform settings (AD7 + AD8)

- Admin user management page (`/utilisateurs`) backed by Better-auth's admin plugin. Filters by role / banned / search; per-row actions for ban (with optional reason + expiration in days), unban, force-logout (revoke all sessions). Self-ban is blocked at the route layer.
 - Platform settings page (`/parametres-plateforme`) edits the two existing `app_config` keys (wallet config and partner revenue rates) via a single `/api/admin/settings` endpoint. After save, the in-memory caches in `wallet.ts` and `partner-revenue-service.ts` are invalidated so live services pick up new values immediately.
 - Three Auth-section open items (admin force-logout / ban / unban UI, team member invite acceptance UI, SMTP credentials in prod) are now satisfied by this work plus the existing `/invite/[id]` route and Gmail App Password setup. Note: production should migrate to a dedicated transactional mail provider before public launch.
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8. Publeader — Audit trail & GDPR (X4)

- Audit log collection + service. Twelve admin mutations now record audit entries (actor, action, target, before/after snapshots, IP): user ban/unban/revoke-sessions, invoice send/mark-paid/delete, validation approve/reject/request-info, withdrawal process/reject, partner payout mark-paid, settings update (with full before/after diff), reports generate/delete, and Stripe webhook processing as a system event.
- Admin `/audit` page: filter by action, actor, date range; expandable row shows the JSON payload (before / after / meta / IP).
- GDPR data export: `/api/me/gdpr/export` builds a ZIP of role-scoped JSON files (user, driver+transactions+withdrawals+vehicles+documents for drivers; company+invoices+campaigns+assets for advertisers; partner+terminals+payouts for partners; plus a manifest).
- GDPR account deletion (anonymization, not hard delete): `/api/me/gdpr/delete` zeroes PII on the user row + role-specific row, randomises the password hash and wipes sessions to block future logins, destroys uploaded

documents and vehicle photos from Cloudinary, and writes a compliance record into a new `gdpr_deletions` collection. Financial records (transactions, withdrawals, invoices) are retained per the FR Code de commerce 10-year obligation, but no longer attributable to the user.

- New `/mes-donnees` page available to all logged-in roles: download ZIP button + delete-account modal that requires typing `SUPPRIMER`.
 - Document retention policy moved to a single source of truth (`lib/retention-policy.ts`) consumed by both the GDPR audit report and this report. No automated purge — admins enforce manually.
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9. Publeader — Advertiser performance + impressions (A6)

- New `lib/campaign-performance-service.ts` exposes two aggregators:
 - `getCompanyPerformance(companyId, period)` — portfolio-wide KPIs (impressions total, reach via distinct terminals, lifetime km, campaign-days = active days × drivers/terminals), daily impressions timeline, top-6 cities by impressions, and a top-6 campaign share with a tail bucketed into "Autres".
 - `getCampaignPerformance(companyId, campaignId, period)` — same KPI block scoped to one campaign + fill rate (drivers vs needed for floorage, impressions vs target for borne) and budget-consumed %.
 - Periods: 7d / 30d / 90d / 365d. Endpoints:
`GET /api/me/performance?period=...` + `GET /api/me/campaigns/[id]/performance?period=...`
Both `requireAdvertiser`-gated; per-campaign route checks ownership.
 - Web `EnterprisePerformance` page rewritten — period switch refetches, KPI grid + impressions area chart + per-campaign share + city bars all live. The campaign-detail overview tab gains a "Performance — 30 j" panel with 4 KPI tiles, sparkline, and fill-rate progress.
 - Mobile advertiser `stats.tsx` rewritten with period pills and real KPIs / bar chart (last 14 buckets) / top-cities / top-campaigns, backed by a new typed client at `lib/performance-api.ts`.
 - Reach + hours from the original spec map to "unique terminals" and "campaign-days" since the data model lacks viewer + hour tracking; all four KPIs use only data we actually capture.
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10. Publeader — Advertiser billing, payment methods & balance (A7)

- `CompanyDoc` extended with an optional `billing` block (email, address, note) and a `stripeCustomerId` field set lazily on first Checkout / Portal call. Legacy companies don't need a backfill.
- New `lib/billing-service.ts`:
 - `ensureStripeCustomer(companyId)` — idempotent. Creates the Stripe Customer with company name + billing email + EU VAT id and stores the id on the doc.

- `createPortalSession(companyId, returnUrl)` — wraps `stripe.billingPortal.sessions.create` so the advertiser manages cards on Stripe-hosted pages (no in-app Elements, mirrors the AD3 Checkout pattern).
- `listSavedPaymentMethods(companyId)` — pulls cards live from Stripe plus the customer's `default_payment_method`.
- `getBillingMetrics(companyId)` — account balance (open invoices), monthly MRR (paid this month), lifetime spend, overdue subset, and next-due-date.
- `getBillingDashboard(companyId)` — single-call merge of profile + metrics + saved cards.
- Existing AD3 Checkout flow upgraded: when a Stripe customer exists we pass `customer` instead of `customer_email` and add `payment_intent_data.setup_future_usage = "off_session"`, so the first card the advertiser uses gets saved automatically and shows up at the next Checkout — no manual "save card" step needed.
- New routes: `GET /api/me/billing` (dashboard payload), `PUT /api/me/billing` (billing email / address / note with email regex validation, mirrored to Stripe Customer best-effort), and `POST /api/me/billing/portal` (returns the hosted portal URL with a sanitised return URL).
- `EnterpriseFacturation` rewritten end-to-end: 4-KPI banner (solde dû, MRR, total dépensé, prochaine échéance), invoice table from `/api/me/invoices` with a Pay button per open row that delegates to the existing `/api/me/invoices/[id]/checkout` endpoint, saved-cards panel + "Gérer" button that redirects to the Stripe Portal, and a billing-profile editor with read-only legal info (legalName, SIRET, VAT — those still live in the A1 company-profile screen).
- Stripe webhooks for payment confirmation + dispute + refund were already delivered with AD3; A7 didn't add new event types.